Introduction to Scope

SESSION OVERVIEW

Replacing hours of manual copy-paste with small LLM agents — and learning LangChain, LangGraph, LangSmith, and FastAPI along the way.

The Problem Today

The Campus-Data researcher team manually fills 5 staging tables by hand:

- staging_company — 163 fields
- staging_skills_topics
- staging_company_skill_levels
- job_role_details_json
- innovx_json

Every new company = hours of copy-paste from web search → spreadsheet → CSV → trigger.

The Solution

We replace that loop with small LLM agents orchestrated through a modern AI stack. Agents handle research, validation, cleaning, and ETL — so researchers focus on what matters.

This is an assistive tool — not a replacement for researchers.

What We Will Build

Milestone 1

4 Research Agents: Researcher, Consolidator, Skill Set, Hiring Process

Milestone 2

Validation + Cleaning + ETL agents orchestrated in LangGraph

Milestones 3 + 4

Two FastAPI microservices: InnovX and Placement Analytics

Milestone 5

A thin frontend plugged into the microservices

What We Will *Not* Build

Scope: 10 Core Fields Only

Full automation of all 163 fields is out of scope — we target 10 core fields to save API credits.

No Production Hardening

Auth, rate limiting, multi-tenant logic, and observability stack are excluded.

No Model Training

Pure prompt + schema engineering only — no custom model training or fine-tuning.

Guardrails

Self-Healing Agents

Generate → Pydantic gate → semantic assertion → bounded retry (≤ 3 tries)

Dual-Mode Execution

Every agent runs manually as a single Python script in Milestone 1, then auto-wired in LangGraph in Milestone 2.

Consolidated Stack

The bug-fix microservice is dropped — its functionality is merged into the Validation/Cleaning/ETL stack.